STATISTICAL FOUNDATIONS FOR MACHINE LEARNING

Notes/Scribes for TA Session - Week 2

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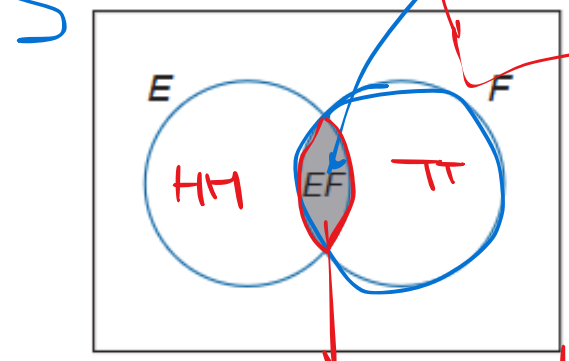
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Conditional Probability

Conditional probability of E given that F has occurred, and is denoted by

$P(E|F)$

$= \frac{P(E \cap F)}{P(F)} = \frac{2}{3}$



$P(F) = 3/4$

$P(E \cap F) = P(HT, TH) = 2/4$

A general formula for $P(E|F)$ that is valid for all events E and F. If the event F occurs, then in order for E to occur it is necessary that the actual occurrence be a point in both E and F; that is, it must be in EF. Now, since we know that F has occurred, it follows that F becomes our new (reduced) sample space and hence the probability that the event EF occurs will equal the probability of EF relative to the probability of F. That is,

$$P(E|F) = \frac{P(EF)}{P(F)}$$

if E and F are independent,

$P(E).P(F) = P(EF)$

$$P(E|F) = \frac{P(EF)}{P(F)}$$

$$P(E|F) = \frac{P(E).P(E)}{P(F)}$$

$P(E|F) = P(E)$

Similarly,

$P(F|E) = P(F)$

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Conditional Probability

$$P(E|F) = \frac{P(EF)}{P(F)}$$

$$P(F|E) = \frac{P(EF)}{P(E)}$$

$$P(F \mid E) = \frac{P(E \mid F)P(F)}{P(E)}$$

Standard Bayes Formula

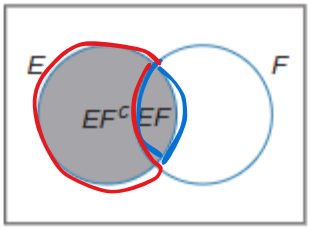
Bayes Formula

Let E and F be events. We may express E as

$$E = EF \cup EF^c$$

As EF and EF^c are clearly mutually exclusive, we have by Axiom 3 that

$$\begin{aligned} P(E) &= P(EF) + P(EF^c) \\ &= P(E|F)P(F) + P(E|F^c)P(F^c) \\ &= P(E|F)P(F) + P(E|F^c)[1 - P(F)] \end{aligned}$$



LAW OF TOTAL PROBABILITY

Equation states that the probability of the event E is a weighted average of the conditional probability of E given that F has occurred and the conditional probability of E given that F has not occurred: Each conditional probability is given as much weight as the event it is conditioned on has of occurring.

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Question:

A card is drawn randomly from a standard deck of 52 playing cards. Given that the card drawn is a **face card**, find the probability that it is a **king**.

Answer:

Let:

- A = card is a King
- B = card is a Face Card

We need:

$$P(A|B)$$

There are 12 face cards: 4 Jacks, 4 Queens, and 4 Kings.

There are 4 Kings.

Using conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Since every King is a face card,

$$A \cap B = A$$

Therefore,

$$P(A|B) = \frac{4/52}{12/52}$$

$$= \frac{4}{12} = \boxed{\frac{1}{3}}$$

Final Answer:

$$\boxed{\frac{1}{3} \approx 33.33\%}$$

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In a class of 40 students, 25 students study Mathematics, 18 students study Physics, and 12 students study both Mathematics and Physics. If a student is selected randomly from those who study Mathematics, what is the probability that the student also studies Physics?

- M = student studies Mathematics
- P = student studies Physics

We are given:

$$P(M) = \frac{25}{40}$$

and

$$P(M \cap P) = \frac{12}{40}$$

We need:

$$P(P|M)$$

Using conditional probability:

$$P(P|M) = \frac{P(P \cap M)}{P(M)}$$

Substitute:

$$P(P|M) = \frac{12/40}{25/40}$$

$$= \frac{12}{25}$$

Therefore,

$$\boxed{P(P|M) = \frac{12}{25} = 0.48}$$

Final Answer:

$$\boxed{48\%}$$

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25. Assume a certain city X experience a hot day day with probability 0.4 and a cold day with probability 0.6. Each hot day experiences rain with probability 0.2 while a cold day experiences rain with probability 0.7. Given a particular day is rainy, what is the probability that the day is hot.

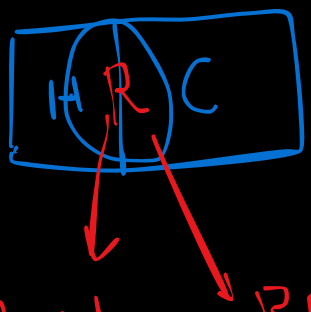
- H = the day is **hot**
- C = the day is **cold**
- R = the day is **rainy**

$(R|H) = 0.2$ $P(H|R) = ?$
$(R|C) = 0.7$

Given:

$$P(H) = 0.4, \qquad P(C) = 0.6$$

$$P(R|H) = 0.2, \qquad P(R|C) = 0.7$$



$R \cap H$ $R \cap C$

We want:

$$P(H|R) = \frac{H \cap R}{R}$$

$= (R|H) \cdot P(H) +$

Step 1: Apply Bayes' theorem

$$P(H|R) = \frac{P(R|H)P(H)}{P(R)}$$

$= \frac{0.08}{0.5} = \frac{8}{50}$
$= 0.16$

First find $P(R)$ using the law of total probability:

$$P(R) = P(R|H)P(H) + P(R|C)P(C)$$

$$= (0.2)(0.4) + (0.7)(0.6)$$

$$= 0.08 + 0.42 = 0.50$$

Therefore,

$$P(H|R) = \frac{(0.2)(0.4)}{0.50}$$

$$= \frac{0.08}{0.50} = \boxed{0.16}$$

Final Answer

$$\boxed{P(\text{Hot} \mid \text{Rainy}) = 0.16 = 16\%}$$

Interpretation: Even though hot days occur 40% of the time, once we know it is raining, the probability that it is hot drops to **16%**, because rain is much more likely on cold days.

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30. A man is known to speak the truth 3 out of 4 times. He throws a die and reports that it is a six. Find the probability that it is actually a six.

Let:

- S = die actually shows a six
- R = man reports that it is a six

We need:

$$P(S|R)$$

Given

The man tells the truth 3 out of 4 times:

$$P(R|S) = \frac{3}{4}$$

If the die is **not** a six, then he lies with probability:

$$1 - \frac{3}{4} = \frac{1}{4}$$

Thus,

$$P(R|S^c) = \frac{1}{4}$$

Since the die is fair:

$$P(S) = \frac{1}{6}, \qquad P(S^c) = \frac{5}{6}$$

Apply Bayes' theorem

$$P(S|R) = \frac{P(R|S)P(S)}{P(R|S)P(S) + P(R|S^c)P(S^c)}$$

Substitute:

$$P(S|R) = \frac{\frac{3}{4} \cdot \frac{1}{6}}{\left(\frac{3}{4} \cdot \frac{1}{6}\right) + \frac{1}{4} \cdot \frac{5}{6}}$$

$$= \frac{\frac{1}{8}}{\frac{1}{8} + \frac{5}{24}}$$

$$= \frac{\frac{3}{8}}{\frac{3}{8} + \frac{5}{8}} = \boxed{\frac{3}{8}}$$

Therefore,

$$\boxed{P(\text{actually a six} \mid \text{reported six}) = \frac{3}{8} = 37.5\%}$$

Answer: $\boxed{37.5\%}$.

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Example 1.26 (False positive paradox [5])

A certain disease affects about 1 out of 10,000 people. There is a test to check whether the person has the disease. The test is quite accurate. In particular, we know that

- the probability that the test result is positive (suggesting the person has the disease), given that the person does not have the disease, is only 2 percent;
- the probability that the test result is negative (suggesting the person does not have the disease), given that the person has the disease, is only 1 percent.

A random person gets tested for the disease and the result comes back positive. What is the probability that the person has the disease?

$(+|D^c) = 0.02$

$(-|D) = 0.01$

$(D|+)$

$$P(D) = 1/10000 = 0.0001$$
$$P(+ \mid D) = 0.99 \text{ (sensitivity)}$$
$$P(+ \mid D^c) = 0.02 \text{ (false positive rate)}$$
$$P(D^c) = 0.9999$$

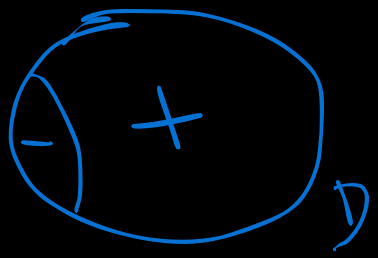
$P(-|D) = 0.01$

$\therefore P(+|D) = [1 - P(-|D)]$

from Bayes Formula

$$P(E) = P(EF) + P(EF^c)$$
$$= P(E|F)P(F) + P(E|F^c)P(F^c)$$

$$P(+) = P(+ \mid D)P(D) + P(+ \mid D^c)P(D^c).$$



Standard Bayes Formula

$$P(F \mid E) = \frac{P(E \mid F)P(F)}{P(E)}$$

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+)}$$

$$P(D \mid +) = \frac{P(+ \mid D)P(D)}{P(+ \mid D)P(D) + P(+ \mid D^c)P(D^c)}.$$

Bayes:

$$\Pr(D \mid +) = \frac{\Pr(+ \mid D)\Pr(D)}{\Pr(+ \mid D)\Pr(D) + \Pr(+ \mid \bar{D})\Pr(\bar{D})} = \frac{0.99 \cdot 0.0001}{0.99 \cdot 0.0001 + 0.02 \cdot 0.9999}$$

Compute the denominator:

$$0.99 \cdot 0.0001 = 0.000099, \qquad 0.02 \cdot 0.9999 = 0.019998,$$

so

$$\Pr(D \mid +) = \frac{0.000099}{0.000099 + 0.019998} = \frac{0.000099}{0.020097} \approx 0.00493.$$

Answer: $\boxed{\approx 0.493\%}$ (about 1 in 203).

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